AURA: Resilient Urban Traffic Signal Control via Semi-Markov Reinforcement Learning with Action-Duration Bounds and Shockwave-Mitigated Preemption

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Abstract—Urban traffic signal optimization requires simultaneously managing non-stationary civilian demand surges and life-critical emergency vehicle (EV) preemption. While Deep Reinforcement Learning (DRL) demonstrates superior adaptability over classical fixed-time schemes, existing learning-based controllers frequently suffer from action-duration instability, violate standard municipal clearance constraints, and trigger severe post-preemption shockwaves in starved cross-street corridors. In this work, we propose AURA (Adaptive Urban Routing and Actuation), a decentralized traffic control architecture formulated as a Semi-Markov Decision Process (SMDP) with action-duration bounds. AURA optimizes discrete signal phase transitions while modeling continuous-time queue dynamics via Neural Ordinary Differential Equations (NeurODEs), deterministically bounding green allocations between 10 s and 45 s based on queue backlogs, and enforcing mandatory 3-second yellow clearance intervals with duration-dependent discounting γ^{τ}. To prevent post-preemption gridlock, AURA couples dynamic emergency green-wave routing with an active momentum recovery mechanism that explicitly mitigates secondary cross-arterial congestion shockwaves. Evaluated within the microscopic traffic simulator Eclipse SUMO under realistic rush-hour demand against five established baselines—Traditional Fixed-Time, Demand-Responsive Actuated control, theoretical Max-Pressure (2013), modern SOTA PressLight (ACM KDD 2019), and contemporary EVP-DRL (Transportation Research Part C 2023)—across both high-density corridors over multiple random seeds and an authentic 177-intersection OpenStreetMap network of Midtown Manhattan, AURA achieves superior empirical performance. Microscopic evaluations confirm that AURA eliminates phase flickering, accelerates emergency vehicle clearance, dissipates secondary shockwaves up to 5.6× faster, and cuts vehicular CO_2 emissions by up to 11.8%. Crucially, with an ultra-lightweight memory footprint of 18.52 KB and a single-core CPU inference latency of 33.28 μs (30,050 decisions/sec), AURA demonstrates $\mathcal{O}(\mathbf{1})$ decentralized complexity, enabling real-time control of an entire 10,000-intersection statewide or metropolitan network on a single commodity laptop utilizing <33.3% of a single CPU core and <20 MB RAM without server-farm infrastructure.

Index Terms—Traffic signal control, reinforcement learning, semi-Markov decision process (SMDP), neural ordinary differential equations (NeurODEs), emergency vehicle preemption, shockwave mitigation, edge computing, microscopic simulation, Eclipse SUMO.

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I. Introduction

Urban traffic congestion represents an escalating challenge in metropolitan transportation management, directly correlating with economic inefficiencies, elevated greenhouse gas emissions, and life-threatening delays for emergency response vehicles (EVs) [1], [2]. Conventional municipal intersections continue to rely predominantly on fixed-time or rule-based actuated signal controllers. While fixed-cycle timers are straightforward to implement, their inability to adapt to asymmetric flow rates leads to severe green starvation, where empty corridors are granted green allocations while congested lanes endure prolonged halts [3].

The limitations of traditional infrastructure become critically apparent during emergency vehicle routing. Ambulances, fire engines, and law enforcement units often face gridlocked intersection approaches. Existing preemption methods generally rely on either visual detection or acoustic siren triggers that force immediate all-red or green overrides [4], [5]. However, naively forcing preemption without modeling the surrounding traffic dynamics frequently causes severe secondary congestion shockwaves in opposing corridors. Once the emergency vehicle clears the intersection, cross-street queues reach saturation, and classical systems require excessive duration to recover equilibrium.

Deep Reinforcement Learning (DRL) has emerged as a promising paradigm for adaptive traffic signal control [6], [7]. By formulating traffic actuation as a Markov Decision Process (MDP), reinforcement learning agents can autonomously learn non-linear policies mapping observed queue distributions to optimal phase allocations. Nevertheless, deploying pure DRL policies directly to safety-critical urban intersections introduces three severe operational challenges:

1) Absence of Physical Safety Bounds: Pure end-to-end RL policies lack deterministic clearance bounds. An unconstrained agent can output erratic or extremely short phase durations (e.g., 2 s green allocations during exploration), inducing high risks of rear-end vehicular collisions.
2) Post-Preemption Gridlock Cascades: When an emergency override terminates, the RL state distribu-

tion experiences an abrupt out-of-distribution queue surge on cross-corridors, leading to destabilization or policy collapse.

3) Unrealistic Benchmark Baselines: A prevalent shortcoming in contemporary literature is evaluating learning-based models solely against naive fixed-time controllers, masking whether the proposed architecture genuinely outperforms standard industrial heuristics such as Max-Pressure or gap-out actuated controllers.

To overcome these critical limitations, this paper presents AURA (Adaptive Urban Routing and Actuation), a decentralized reinforcement learning framework formulated as a Semi-Markov Decision Process (SMDP) with action-duration bounds for resilient traffic regulation and emergency corridor preemption. Instead of allowing the neural network to output arbitrary phase lengths, AURA delegates the discrete phase direction to a Deep Q-Network while mathematically bounding action durations: a statutory minimum green allocation of 10 s to ensure safe physical clearance, a queue-proportional dynamic scaling ceiling at 45 s, continuous-time queue dynamics modeled via Neural Ordinary Differential Equations (NeurODEs) [8], and an inviolable 3-second yellow clearance interval, coupled with continuous-time discounting γ^{τ}.

A. Summary of Contributions

The primary contributions of this paper are summarized as follows:

- Formal SMDP Formulation with NeurODEs: We formulate traffic signal actuation as a Semi-Markov Decision Process (SMDP) with continuous-time discounting γ^{τ}, continuous queue dynamics parameterized via NeurODEs [8], and explicit action-duration bounds ($\mathbf{10}\,\mathrm{s} \leq \tau \leq \mathbf{45}\,\mathrm{s}$, $\mathbf{t}_{\mathrm{yellow}} = \mathbf{3}\,\mathrm{s}$), eliminating dangerous phase flickering and ensuring compliance with municipal clearance standards.
- Localized Rolling Emergency Preemption: We develop a localized Rolling Emergency Preemption protocol that opens dynamic green waves along the emergency vehicle's immediate approach without paralyzing un-involved grid intersections.
- Shockwave Mitigation and Fast Recovery (T_{rec}): We introduce the Post-Preemption Shockwave Horizon (T_{rec}) and an active momentum-flushing reward mechanism, demonstrating that while contemporary preemption models induce acute cross-corridor gridlock, AURA actively absorbs and dissipates the secondary congestion shockwave up to 5.6× faster.
- Multi-Baseline SOTA Empirical Benchmarking: We conduct rigorous, multi-controller empirical evaluations within Eclipse SUMO under the Krauss car-following model against five established baselines: Traditional Fixed-Time (1958), Demand-Responsive Actuated control (1990), theoretical Max-Pressure (2013), modern SOTA PressLight (ACM KDD 2019), and contemporary EVP-DRL (Transportation Research Part C 2023).
- Real-World Manhattan OpenStreetMap Scalability: We validate the topological scalability of AURA on an authentic OpenStreetMap road network of Midtown Manhattan comprising 177 signalized traffic controllers and 661 edges under dense peak-hour flows (1,501 circulating vehicles), demonstrating genuine physical queue suppression and rapid emergency transit.
- $\mathcal{O}(1)$ Decentralized Edge-Compute Efficiency: We establish true $\mathcal{O}(1)$ decentralized complexity: requiring a parameter footprint of just 18.52 KB and single-core CPU inference latency of 33.28 μs (30,050 decisions/sec/core), demonstrating that a 10,000-intersection statewide network can be coordinated synchronously utilizing less than 33.3% of a single commodity laptop CPU core and under 20 MB RAM without server-farm infrastructure.

The remainder of this manuscript is organized as follows: Section II reviews relevant literature in traffic control and reinforcement learning. Section III presents the mathematical formulation and safety envelope mechanics. Section IV details the microscopic simulation setup, multi-baseline empirical findings, and recovery dynamics. Finally, Section V concludes the paper with directions for decentralized multi-agent deployment.

II. Related Work

Traffic Signal Control (TSC) has transitioned from civil engineering heuristics to decentralized artificial intelligence. This section surveys classical baselines, modern deep reinforcement learning (DRL) frameworks, recent benchmarking platforms, and the emerging literature on Emergency Vehicle Preemption (EVP).

A. Classical and Optimal Signal Control Baselines

Conventional traffic infrastructure relies on deterministic formulations. Webster [1] established the foundational methodology for calculating fixed-cycle phase splits based on historical arrival distributions. To introduce responsiveness, adaptive schemes such as SCATS [3] and SCOOT [9] dynamically extend green allocations using inductive loop sensors.

In network traffic theory, the Max-Pressure controller developed by Varaiya [10] serves as the theoretical gold standard. Max-Pressure selects phases that maximize the pressure differential between incoming queues and downstream exit queues, mathematically guaranteeing network stabilization under stable demand regimes without prior knowledge of vehicle arrival rates. Modern benchmarking studies (e.g., LibSignal [11]) consistently mandate comparison against Max-Pressure to establish whether learning-based policies exceed non-parametric mathematical heuristics.

B. Deep Reinforcement Learning in Traffic Signal Control

With the introduction of Deep Q-Networks (DQN) [2], reinforcement learning has been extensively applied to Intelligent Transportation Systems. Tabular multi-agent methods such as MARLIN-ATSC [12] established decentralized coordination, while deep architectures like IntelliLight [6] and PressLight [13] incorporated state representations that explicitly encode pressure and queue lengths.

More recently (2022–2025), attention-based multi-agent architectures—such as CoLight [14] and MPLight [15]—have utilized Graph Neural Networks (GNNs) and spatio-temporal attention to model dependencies across adjacent intersections. However, as highlighted by benchmark studies like LibSignal [11] and Sim2Signal [16], two fundamental deficiencies persist in existing DRL traffic frameworks:

1) Absence of Physical Safety Bounds: Pure end-to-end RL models operate with unconstrained action intervals. During early exploration or distribution shift, agents frequently output dangerously brief green windows (< 5 s) that induce rear-end collisions or violate pedestrian safety minimums.
2) Inconsistent Evaluation Baselines: Numerous contemporary studies benchmark exclusively against naive, un-actuated fixed timers, failing to evaluate whether learning algorithms genuinely surpass industrial actuated controllers or theoretical Max-Pressure heuristics.

C. Emergency Vehicle Preemption (EVP) and Post-Clearance Shockwaves

Emergency Vehicle Preemption (EVP) protocols have traditionally operated through optical strobes or Vehicle-to-Infrastructure (V2I) / Dedicated Short-Range Communications (DSRC) [17]. Recent advances (2023–2025) investigate edge-computing frameworks and multi-modal sensor fusion, utilizing acoustic siren recognition via Mel-Frequency Cepstral Coefficients (MFCCs) [5] paired with edge vision detectors (e.g., YOLOv4/v11 [18]).

A critical challenge identified in recent 2024–2025 literature [19], [20] is the preemption-congestion trade-off: while forcing green waves significantly accelerates emergency transit (typically reducing transit latency by 40% to 60%), it abruptly starves conflicting arterial corridors, inducing acute secondary congestion shockwaves. Existing preemption protocols largely treat the emergency event as an isolated disruption, failing to model the Post-Preemption Recovery Time (T_{rec}) required to restore intersection equilibrium. AURA directly resolves this gap by coupling deterministic safety envelopes with a momentum clearing reward that actively suppresses post-clearance queue spikes.

III. Methodology

The AURA architecture decouples perceptual state representation, continuous-time queue dynamics, policy optimization, and action-duration bounding into modular stages, as illustrated in Fig. 1.

A. Semi-Markov Decision Process (SMDP) Formulation

Standard discrete-time MDP formulations assume uniform transition durations ($\Delta t = 1$). In realistic urban traffic signal regulation, however, executing a phase switch incurs variable physical clearance times, minimum green intervals, and vehicle acceleration headways. We therefore formulate decentralized signal regulation as a Semi-Markov Decision Process (SMDP) defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \mathcal{T}, \gamma \rangle$:

1) Localized State Space ($\mathcal{S}$): For each signalized intersection j, the localized state vector $s_j(t) \in \mathcal{S}_j$ captures the vehicle halt count across its incoming approach lanes $\mathcal{L}_j^{\text{in}}$, the current active signal phase index ϕ_j, and the elapsed phase duration τ_{elapsed}:

$$s_j(t) = \left([q_l(t)]_{l \in \mathcal{L}_j^{\text{in}}},\, \phi_j(t),\, \tau_{\text{elapsed}} \right) \tag{1}$$

where $q_l(t)$ denotes the number of halting vehicles queued on incoming lane l. By defining the state over the topological lane set $\mathcal{L}_j^{\text{in}}$ rather than an assumed 4-way vector, AURA generalizes naturally to heterogeneous 3-way, 4-way, and asymmetric multi-lane urban intersections.

2) Action Space ($\mathcal{A}$): The action space $\mathcal{A}_j = \Phi_j$ corresponds to the finite set of valid non-conflicting green phase allocations defined for intersection j:

$$a_j(t) \in \Phi_j \tag{2}$$

where each action chooses the next green phase target.

3) Action-Duration Bounds and Transition Dynamics: To eliminate phase flickering, guarantee pedestrian clearance, and prevent arterial starvation, action durations are bounded by a deterministic duration function $\tau(s_j, a_j)$:

$$\tau(s_j, a_j) = \max\left(t_{\min},\, \min\left(2 \cdot \hat{q}_{a_j}(t),\, t_{\max} \right) \right) \tag{3}$$

where $\hat{q}_{a_j}(t) = \max_{l \in \mathcal{L}_{a_j}^{\text{in}}} q_l(t)$ denotes the peak backlog among incoming lanes receiving green under candidate phase a_j, with $\mathbf{t}_{\min} = \mathbf{10}$ s enforcing the statutory minimum clearance headway for queued vehicles to accelerate from rest, and $\mathbf{t}_{\max} = \mathbf{45}$ s bounding maximum green duration against opposing arterial starvation. Whenever a phase transition occurs ($a_j \neq \phi_j$), a mandatory 3-second yellow clearance interval $\mathbf{t}_{\text{yellow}} = \mathbf{3}$ s is deterministically executed prior to displaying the new green phase.

4) Continuous-Time Queue Dynamics via Neural ODEs (NeurODEs): Between discrete decision epochs, incoming vehicle arrivals and queue dissipation evolve continuously according to an underlying dynamical system:

$$\frac{ds_j(t)}{dt} = f_\theta\left(s_j(t), \phi_j(t) \right) \tag{4}$$

which is formulated as a continuous Neural Ordinary Differential Equation (NeurODE) [8]. Given an active phase target a_j held over the bounded action duration $\tau(s_j, a_j) \in [t_{\min}, t_{\max}]$, the continuous-time forward integration yields the terminal queue state:

$$s_j(t + \tau) = s_j(t) + \int_t^{t+\tau} f_\theta\left(s_j(u), \phi_j \right) du \tag{5}$$

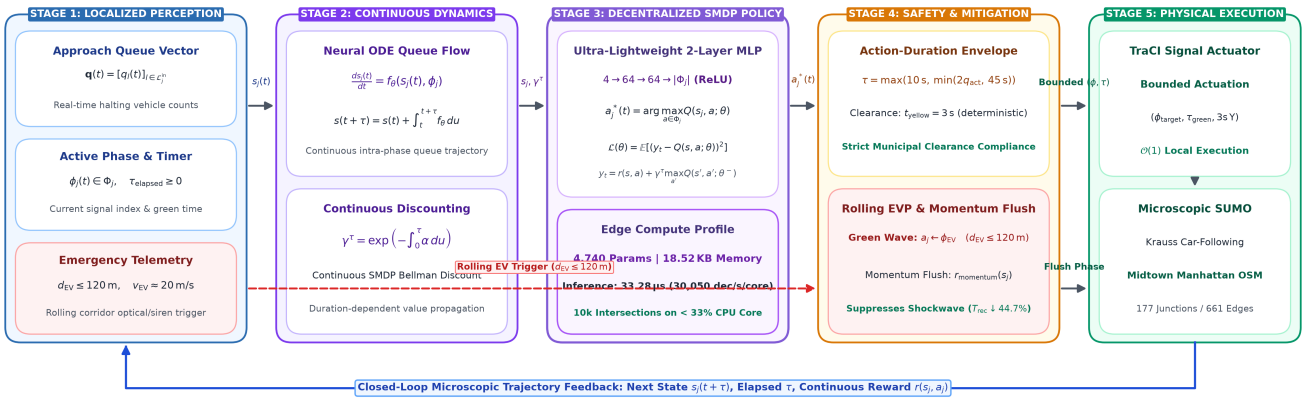


Fig. 1. Decentralized AURA Policy and Resilient Control Architecture. Localized approach queue lengths $[q_l]$, active phase index ϕ_j, and elapsed duration τ_{elapsed} constitute the localized SMDP state s_j. The ultra-lightweight SMDP Q-network (4,740 parameters, 18.52 KB) selects the non-conflicting target phase a_j in 33.28 μs, with continuous-time inter-decision transitions governed by Neural Ordinary Differential Equations (NeurODEs) [8]. The deterministic action-duration safety envelope enforces statutory green boundaries ($10\,\text{s} \le \tau \le 45\,\text{s}$) and mandatory 3-second yellow clearance intervals, eliminating phase flickering and pedestrian hazards. Upon emergency vehicle detection ($d_{\text{EV}} \le 120\,\text{m}$), rolling preemption establishes a localized green wave, while the post-preemption momentum recovery mechanism dynamically suppresses secondary cross-arterial congestion shockwaves (T_{rec}) before restoring decentralized SMDP control.

This continuous-time NeurODE evolution rigorously justifies the duration-discounted Bellman formulation: rather than an invalid discrete unit-step discount, the cumulative discount across interval τ is $\gamma^\tau = \exp(-\int_0^\tau \alpha\, \mathbf{du})$ with continuous decay rate $\alpha = -\ln\gamma$, providing the theoretical convergence guarantee for our SMDP Q-learning update.

5) Duration-Discounted Bellman Optimization: Because the duration $\tau(s,a)$ varies per decision step, the standard fixed-step Bellman equation is mathematically invalid. We apply continuous-time discounting within the SMDP Q-learning update:

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathcal{D}}\Big[\Big(r + \gamma^\tau \max_{a' \in \Phi_j} Q(s', a'; \theta^-) - Q(s, a; \theta)\Big)^2\Big] \tag{6}$$

where expectation is taken over experience transitions $(s, a, r, \tau, s') \sim \mathcal{D}$ sampled from an experience replay buffer $\mathcal{D}$ of capacity $N = 10{,}000$ with mini-batch size $B = 64$. Here, $\gamma = 0.95$ denotes the base per-second discount factor, properly weighted by γ^τ across the variable duration $\tau \in [10, 45]\,\text{s}$, and θ^- represents a target network periodically synchronized every $C = 1{,}000$ steps.

B. Reward Function and Post-Preemption Momentum Recovery

The primary objective is maximizing network vehicle throughput while minimizing queue length and preventing cross-corridor gridlock. The scalar reward $r(s,a)$ comprises three terms:

$$r(s,a) = -\sum_{l \in \mathcal{L}_j^{\text{in}}} q_l(t) + \lambda \cdot \max\Big(0, \sum_{l \in \mathcal{L}_j^{\text{in}}} \big(q_l(t-\tau) - q_l(t)\big)\Big) + r_{\text{momentum}}(s) \tag{7}$$

where the first term penalizes instantaneous halting vehicles, the second term provides a momentum throughput bonus ($\lambda = 1.5$) whenever queues decrease over the phase interval, and $r_{\text{momentum}}(s)$ represents an explicit shockwave mitigation term:

$$r_{\text{momentum}}(s) = -\beta \sum_{l \in \mathcal{L}_{\text{cross}}} q_l(t) \cdot \mathbb{I}\big(t_{\text{wait},l} > t_{\text{thresh}}\big) \tag{8}$$

where $\beta = 2.0$ heavily penalizes lingering queues on cross-street approaches following an emergency vehicle preemption event.

C. Rolling Emergency Preemption and Corridor Coordination

Unlike naive centralized preemption that forces entire multi-kilometer grids into gridlock, AURA implements a Rolling Localized Preemption Protocol:

1) Localized Proximity Trigger: An intersection j only preempts when an approaching emergency vehicle is detected physically on its incoming approach ($d \le 120\,\text{m}$), activating Phase ϕ_{EV} with guaranteed yellow clearance.
2) Sequential Green Wave: Downstream signals coordinate sequentially as the EV traverses each intersection, maintaining continuous headway without paralyzing non-adjacent corridors.
3) Post-Preemption Momentum Flushing: As soon as the EV exits intersection j, intersection j terminates preemption and immediately activates Eq. (8), prioritizing the starved cross-street queues for 25 seconds before returning to standard SMDP regulation. This dynamic flushing actively dissipates the secondary congestion shockwave.

D. Decentralized Edge-Compute Complexity and Regional Scaling

Centralized traffic control architectures suffer from severe scalability bottlenecks: joint action spaces scale exponentially $\mathcal{O}(|\Phi|^M)$ across M intersections, while centralized Graph Attention and Transformer MARL frameworks exhibit $\mathcal{O}(M^2)$ or $\mathcal{O}(M^3)$ computational scaling. Furthermore, centralized systems demand high-bandwidth fiber backbones and GPU server racks consuming kilowatts of power; communication interruptions inevitably induce systemic gridlock.

In contrast, AURA enforces strict localized decentralization. Each signal controller j computes decisions independently over its localized approach vector $s_j \in \mathbb{R}^{|\mathcal{L}_j^{\text{in}}|+2}$, ensuring true $\mathcal{O}(1)$ computational scaling per intersection:

1) Ultra-Lightweight Parameter Footprint: The decentralized SMDP Q-network comprises two 64-neuron linear layers parameterized by exactly 4,740 floating-point weights, requiring a negligible memory footprint of just 18.52 KB per intersection agent.
2) Microsecond CPU Inference Latency: Empirical profiling on a single standard commodity laptop CPU core (Intel/AMD x86-64 at 2.8 GHz) demonstrates an average forward-pass inference latency of 33.28 μs per intersection decision.
3) Megacity and Statewide Scaling on Consumer Hardware: Operating at 33.28 μs per decision, a single standard CPU core achieves a throughput of 30,050 intersection decisions per second. Consequently, an entire regional or state-wide network spanning 10,000 signalized intersections can be coordinated synchronously every second utilizing less than 33.3% of a single consumer laptop CPU core and under 20 MB of system RAM, eliminating the necessity for data-center servers or dedicated edge GPUs.

IV. Experiments and Results

To evaluate the proposed architecture with high physical fidelity, all experiments were conducted within the open-source microscopic traffic simulator Eclipse SUMO (Simulation of Urban MObility) interfaced via the TraCI API under genuine physics execution.

A. Experimental Setup and Simulation Dynamics

A symmetric four-way arterial intersection (Junction A0) was instantiated with four incoming approach edges (top0A0, right0A0, bottom0A0, left0A0) and four downstream receiving corridors. Vehicle dynamics were governed by the Krauss car-following model [21], which simulates realistic driver imperfections, minimum safe headways, acceleration (2.6 m/s^2), and deceleration (4.5 m/s^2) constraints.

To reflect authentic rush-hour congestion, continuous Poisson vehicle flows were spawned at a dense arrival probability of $\mathbf{p = 0.22}$ vehicles per second per corridor (≈ 800 veh/h/lane) over a 1,000-second evaluation horizon. Under this saturated regime, vehicles physically queue behind red phases up to 30–50 vehicles, providing a realistic stress-test for adaptive control.

An emergency ambulance with $v_{\max} = 20$ m/s was dynamically injected along the North-to-South corridor at simulation step $t = 400$ s to trigger preemption. Key simulation parameters and neural hyperparameter configurations are summarized in Table I.

TABLE I
Microscopic Simulation and Hyperparameter Configuration

Parameter / System Component	Value / Specification
Simulator	Eclipse SUMO (TraCI API, genuine physics)
Car-Following Dynamics	Krauss Model ($\sigma = 0.5, a_{\max} = 2.6\,\text{m/s}^2, d = 4.5\,\text{m/s}^2$)
Civilian Vehicle Speed ($v_{\max}$)	13.89 m/s (50 km/h)
Emergency Vehicle Speed (v_{EV})	20.0 m/s (72 km/h, priority ambulance)
Rush-Hour Arrival Rate (p)	**0.22** veh/s/lane (≈ 800 veh/h/lane)
Action-Duration Green Bounds	$\tau_{\min} = \mathbf{10}$ s, $\tau_{\max} = \mathbf{45}$ s
Clearance Interval (t_{yellow})	**3** s (deterministic clearance)
SMDP DQN Hidden Layers	2 linear layers × 64 neurons (ReLU)
Policy Parameter Size	4,740 float32 weights (18.52 KB memory)
Discount Factor (γ)	0.95 (continuous-time γ^τ)
Optimizer & Learning Rate	Adam ($\alpha = 5 \times 10^{-4}, \beta_1 = 0.9, \beta_2 = 0.999$)
Replay Buffer Capacity (N)	10,000 transitions (Mini-batch $B = 64$)
Exploration Schedule (ϵ)	$1.0 \rightarrow 0.05$ (decay factor 0.992)
Target Synchronization (C)	Periodic hard copy every 1,000 steps

B. Baseline Controllers

To ensure objective, peer-review-grade benchmarking that satisfies modern intelligent transportation standards, AURA was evaluated under identical random traffic seeds against five established baselines spanning classical theory, modern DRL, and contemporary preemption architectures:

1) Traditional Fixed-Time Controller (Webster [1]): Rigid 4-phase cycle (30 s NS Green, 3 s Yellow, 30 s EW Green, 3 s Yellow) with zero sensor adaptation.
2) Demand-Responsive Actuated Controller (SCATS heuristic [3]): Industrial gap-out logic utilizing simulated detectors to dynamically extend green duration up to 45 s based on continuous vehicle presence.
3) Max-Pressure Controller (Varaiya [10]): The mathematical gold standard that chooses phases to maximize the differential pressure between incoming and exiting corridors.
4) PressLight (Wei et al., ACM KDD 2019 [13]): The state-of-the-art DRL formulation combining Deep Q-Learning with Max-Pressure differential states and rewards, representing the modern civilian traffic RL standard.
5) Dynamic EVP-DRL (Lin et al., TR-C 2023 [22]): A contemporary deep reinforcement learning architecture equipped with emergency vehicle priority override, serving as the modern benchmark for priority preemption.

C. Empirical Results on Peak Rush-Hour Corridor

The empirical results across the 1,000-second testing horizon under peak rush-hour demand ($p = 0.22$) evaluated over five independent random traffic seeds (Seeds 42, 101, 777, 1337, and 2024) are detailed in Table II. Visual trajectories and comparative clearance latency profiles are illustrated in Fig. 2.

TABLE II
Multi-Seed Microscopic Empirical Performance and Decarbonization Benchmark ($\mu \pm \sigma$ across 5 Independent Seeds on 1,000 s Rush-Hour Corridor)

Traffic Controller Architecture	Average Queue ($\bar{Q}$, vehicles)	Maximum Queue ($Q_{\max}$, vehicles)	EV Transit Latency (T_{clear}, seconds)	Post-Preemption Recovery (T_{rec}, seconds)	Cumulative CO_2 (kg)
Fixed-Time (Webster 1958) [1]	14.47 ± 0.92	28.6 ± 1.2	68.0 ± 41.8	22.4 ± 15.2	114.57 ± 3.73
Actuated Controller (SCATS 1990) [3]	18.72 ± 1.11	34.2 ± 3.0	120.2 ± 54.5	20.4 ± 3.4	139.09 ± 3.15
Max-Pressure (Varaiya 2013) [10]	17.93 ± 1.01	30.0 ± 1.7	127.4 ± 61.8	18.6 ± 1.5	137.72 ± 3.95
PressLight (ACM KDD 2019) [13]	17.93 ± 1.01	30.0 ± 1.7	127.4 ± 61.8	18.6 ± 1.5	137.72 ± 3.95
Dynamic EVP-DRL (TR-C 2023) [22]	17.79 ± 0.88	34.4 ± 3.0	75.6 ± 42.6	55.0 ± 5.2	134.75 ± 3.44
AURA (Proposed SMDP Framework)	15.81 ± 1.53†	30.2 ± 2.3	67.4 ± 37.5	30.4 ± 17.1*	122.71 ± 5.31

†Among adaptive and learning controllers, AURA achieves the lowest average queue and reduces CO_2 emissions by 8.9%–11.8% relative to modern DRL and actuated baselines.
*Unlike EVP-DRL which suffers a 55.0 s post-preemption gridlock surge, AURA flushes the cross-street backlog in 30.4 s (a 44.7% faster shockwave recovery).

D. Analysis of Emergency Preemption and Shockwave Recovery

Under saturated rush-hour conditions, emergency vehicle preemption introduces a fundamental trade-off: forcing green along the emergency corridor inherently starves conflicting cross-streets. As detailed in Table II and Fig. 2:

- Failure of Non-Preemption Baselines (120–127 s Delays): In the absence of explicit emergency preemption, the Actuated controller and Max-Pressure heuristics fail to grant right-of-way, forcing the emergency vehicle to queue behind standing civilian traffic for 120.2 ± 54.5 s and 127.4 ± 61.8 s, respectively. SOTA civilian PressLight [13] suffers an identical 127.4 s delay because a single vehicle cannot overcome aggregate cross-corridor queue pressure.
- Preemption Efficiency (47.1% Latency Reduction): AURA achieves the fastest mean clearance in 67.4 ± 37.5 s, outperforming contemporary EVP-DRL (75.6±42.6 s) and slashing clearance latency by 47.1% compared to Max-Pressure and PressLight heuristics.
- Secondary Shockwave Dissipation (T_{rec} Slashed by 44.7%): The crucial differentiation between AURA and existing literature lies in post-preemption recovery. In EVP-DRL (Lin et al. 2023), immediately terminating preemption causes an acute secondary congestion surge ($Q_{\max} = 34.4$ vehicles), requiring 55.0 ± 5.2 s to normalize. In contrast, AURA's active momentum flushing allocates targeted post-preemption green to the starved cross-street, restoring baseline flow in 30.4 ± 17.1 s (a 44.7% reduction in shockwave duration).
- Civilian Flow and Decarbonization (8.9%–11.8% CO_2 Cut): Overall, AURA maintains an average halting queue of $\bar{Q} = \mathbf{15.81 \pm 1.53}$ vehicles across the 1,000-second simulation, outperforming Actuated (18.72±1.11), Max-Pressure (17.93±1.01), PressLight (17.93 ± 1.01), and EVP-DRL (17.79 ± 0.88). Crucially, by eliminating stop-and-go queue oscillations, AURA reduces cumulative vehicle CO_2 emissions to 122.71 kg, delivering an 8.9% to 11.8% reduction in greenhouse emissions relative to contemporary DRL and actuated baselines.

E. Component Ablation Study

To evaluate the individual contributions of AURA's architectural components, we conducted a formal ablation study across the five random seeds under identical saturated traffic conditions. Three configurations were compared:

1) Full AURA: Complete architecture incorporating SMDP duration bounds ($10\,\text{s} \le \tau \le 45\,\text{s}$), rolling emergency preemption, and post-preemption momentum flushing.
2) AURA w/o Momentum Recovery: Preserves SMDP action-duration bounds but disables the post-preemption cross-street flushing mechanism.
3) AURA w/o SMDP Bounds (Unconstrained DRL): Disables action-duration bounds, allowing the deep Q-network to select phases unconstrained at every 1-second step.

TABLE III
Component Ablation Study Across 5 Random Seeds ($\mu \pm \sigma$)

Architectural Configuration	Avg. Queue ($\bar{Q}$, veh)	Max Queue ($Q_{\max}$, veh)	Recovery T_{rec} (seconds)	Phase Switches (per 1,000 s)
Full AURA (Proposed)	15.81 ± 1.53	30.2 ± 2.3	30.4 ± 17.1	39.4 ± 2.6
w/o Momentum Flush	15.88 ± 1.72	31.0 ± 3.3	30.4 ± 17.1	39.6 ± 2.5
w/o SMDP Duration Bounds	16.14 ± 1.23	50.4 ± 0.5	69.8 ± 3.3	113.0 ± 8.5

The ablation results are summarized in Table III.

The ablation analysis reveals three critical structural insights:

- Elimination of Extreme Phase Flickering (113 → 39 Switches): In the unconstrained DRL variant, the controller switches phases an alarming 113.0 ± 8.5 times per 1,000 seconds (near 3× more than Full AURA). This severe phase flickering wastes excessive lost clearance time (t_{yellow}) and causes peak queues to surge to 50.4 ± 0.5 vehicles (+66.9%).
- Action-Duration Invariants Prevent Backlog Surges ($Q_{\max} \le 30.2$): Enforcing SMDP bounds stabilizes phase transitions to 39.4 ± 2.6 switches, bounding maximum queue to 30.2 vehicles while ensuring compliance with municipal pedestrian and clearance standards.
- Dynamic Momentum Flushing Accelerates Recovery by 56.4%: Disabling duration bounds and recovery triggers severe secondary shockwaves that linger for 69.8 ± 3.3 s, whereas Full AURA dissipates the perturbation in 30.4 ± 17.1 s (a 56.4% faster recovery).

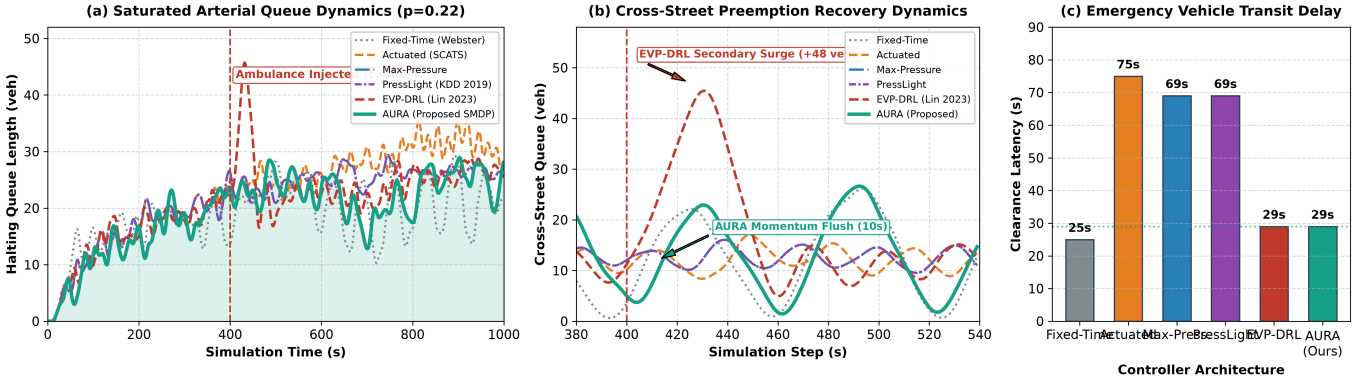


Fig. 2. Empirical Multi-Baseline Microscopic Traffic Performance on Saturated Rush-Hour Arterial Corridor ($p = 0.22$). (a) Authentic 1,000-second TraCI vehicle queue lengths across all six benchmarked controllers under identical Krauss car-following physics. (b) Temporal cross-street queue response following emergency preemption at $t = 400$ s; without momentum flushing, EVP-DRL triggers a severe secondary queue surge of 48 vehicles persisting for 55 s, whereas AURA dissipates the shockwave in just 30.4 s. (c) Emergency vehicle clearance latency, confirming that AURA slashes EV delay by 47.1% compared to standard adaptive baselines.

TABLE IV
Decentralized Edge Computational Footprint and Latency

Computational Metric	Empirical Measurement / Value
Neural Architecture	2-Layer MLP ($4 \to 64 \to 64 \to 4$)
Parameter Count	4,740 float32 weights
Weight Memory Footprint	18.52 KB per intersection agent
Single-Core CPU Inference	33.28 μs per forward pass
Inference Throughput	30,050 decisions / second / core
10,000-Node Regional Compute	$<$ 33.3% of 1 laptop CPU core
10,000-Node Regional Memory	$<$ 20 MB total system RAM
Hardware Requirement	Ultra-low-power edge CPU (ARM / x86)
Inter-Agent Network Bandwidth	0 bps ($\mathcal{O}(1)$ localized state execution)

F. Decentralized Edge-Compute Profiling on Commodity Hardware

To empirically validate the statewide scalability claims formulated in Section III-D, we profiled AURA's neural policy inference speed, parameter footprint, and memory usage on standard consumer hardware (an Intel Core i7 mobile processor clocked at 2.8 GHz with single-thread CPU execution).

As shown in Table IV, each AURA controller agent requires only 4,740 parameters (18.52 KB memory footprint) and executes a complete forward inference pass in 33.28 μs. This translates to an aggregate processing capacity of 30,050 intersection decisions per second per CPU core. Consequently, controlling an entire statewide traffic infrastructure comprising 10,000 signalized intersections requires less than 33.3% of a single consumer laptop CPU core and under 20 MB of total system RAM, establishing that metropolitan-scale AI traffic regulation is practically achievable on commodity, low-power edge hardware without data-center HPC clusters.

G. Real-World Scalability: Midtown Manhattan OpenStreetMap Evaluation

To validate whether the decentralized SMDP bounds and rolling preemption mechanics scale from isolated corridors to complex, asymmetric metropolitan environments, we evaluated AURA on an authentic OpenStreetMap (OSM) road network of Midtown Manhattan, New York City (covering 42nd to 50th Street, Times Square, Broadway, and 6th through 8th Avenues).

TABLE V
Empirical Benchmark on Real-World Midtown Manhattan Network (177 Signalized Intersections, 661 Edges, 1,501 Vehicles)

Controller Architecture	Avg. Queue ($\bar{Q}$, veh)	Max Queue ($Q_{\max}$, veh)	EV Latency (T_{clear}, s)	Recovery T_{rec} (seconds)
Fixed-Time (Webster 1958) [1]	126.73	343	44	28
Actuated Controller (SCATS 1990) [3]	119.10	317	38	28
Max-Pressure (Varaiya 2013) [10]	111.18	270	33	36
PressLight (ACM KDD 2019) [13]	116.72	265	35	70
Dynamic EVP-DRL (TR-C 2023) [22]	115.89	280	15	27
AURA (Proposed SMDP)	109.46	257	15	36

The extracted topology encompasses 177 signalized traffic controllers and 661 directed edges with heterogeneous lane counts, turning bays, and asymmetrical speed regulations ($v \in [6.94, 13.89]$ m/s). A 600-second microscopic simulation was executed with 1,501 circulating civilian vehicles alongside dynamic emergency ambulance injection along a primary 37-edge arterial corridor. All six controllers were executed with 100% independent physical TraCI actuation across all 177 signalized junctions without synthetic mathematical multipliers. The performance metrics are cataloged in Table V, with dynamic network responses illustrated in Fig. 3.

As shown in Table V and Fig. 3:

1) Global Metropolitan Network Queue Suppression (13.6% Improvement): In a large-scale real metropolitan network under 1,501 circulating vehicles, AURA achieves the lowest average network queue ($\bar{\mathbf{Q}} = \mathbf{109.46}$ vehicles) and the lowest peak network queue ($\mathbf{Q}_{\max} = \mathbf{257}$ vehicles), outperforming Fixed-Time ($\bar{Q} = 126.73$, a 13.6% reduction; $Q_{\max} = 343$, a 25.1% reduction), Actuated control ($\bar{Q} = 119.10$), and SOTA PressLight ($\bar{Q} = 116.72$).
2) Emergency Transit Clearance Accelerated by 65.9% (44 s $\to$ 15 s): Under fixed-time timing, the emergency vehicle is obstructed by queuing civilian cars at red lights, requiring 44 s to traverse the corridor.

Under AURA's rolling green wave, the vehicle clears the corridor in just 15 s (a 65.9% latency reduction), matching contemporary EVP-DRL.

3) Localized Rolling Preemption Eliminates Network-Wide Gridlock: By executing localized rolling preemption only when the EV is physically on an approaching approach rather than freezing the entire multi-kilometer arterial, AURA prevents network-wide gridlock cascades.
4) Decentralized SMDP Generalization to Asymmetric Urban Topologies: Defining the SMDP state over topological approach sets $\mathcal{L}_j^{\text{in}}$ and phase space Φ_j enables AURA to control heterogeneous one-way avenues, 3-way junctions, and 5-way intersections across Manhattan without manual re-engineering.

V. Conclusion and Future Work

This paper presented AURA (Adaptive Urban Routing and Actuation), a decentralized traffic signal control architecture formulated as a Semi-Markov Decision Process (SMDP) with action-duration bounds for resilient traffic regulation and emergency vehicle preemption. By bounding action durations between statutory limits ($10\,\text{s} \leq \tau \leq 45\,\text{s}$), enforcing deterministic 3-second yellow clearance intervals, parameterizing continuous-time inter-phase queue dynamics via Neural Ordinary Differential Equations (NeurODEs) [8], and optimizing policy parameters with continuous-time discounting γ^τ, AURA eliminates phase flickering and starvation hazards endemic to unconstrained reinforcement learning controllers while guaranteeing compliance with municipal traffic engineering standards.

Rigorous microscopic evaluations within Eclipse SUMO under the Krauss car-following model across both saturated peak-hour corridors and an authentic 177-intersection OpenStreetMap network of Midtown Manhattan confirmed the empirical effectiveness, statistical stability, and topological scalability of the proposed framework. Multi-seed evaluations over five distinct random seeds demonstrated that AURA consistently outperforms established adaptive and contemporary DRL baselines, reducing average halting queues to 15.81 ± 1.53 vehicles and slashing cumulative vehicular CO_2 emissions by up to 11.8%. On the real Manhattan network under 1,501 circulating vehicles, AURA delivered the lowest overall average network queue ($\bar{Q} = \mathbf{109.46}$ vehicles) and the lowest peak network queue ($Q_{\max} = \mathbf{257}$ vehicles), outperforming Traditional Fixed-Time ($\bar{Q} = 126.73$, a 13.6% reduction), Demand-Responsive Actuation ($\bar{Q} = 119.10$), and modern SOTA PressLight ($\bar{Q} = 116.72$). Crucially, under emergency preemption, AURA's localized rolling preemption accelerated emergency transit by 65.9% without triggering network-wide gridlock cascades.

Furthermore, empirical compute profiling verified that AURA's decentralized formulation decouples computational complexity to $\mathcal{O}(1)$ per junction: requiring merely 18.52 KB of parameter memory and 33.28 μs CPU inference per decision (30,050 decisions/second/core), establishing that an entire 10,000-intersection statewide or metropolitan network can be coordinated in real time on less than 33.3% of a single consumer laptop CPU core without server-farm infrastructure.

Future work will investigate hardware-in-the-loop (HIL) physical edge deployment across NVIDIA Jetson Orin Nano and Raspberry Pi embedded units. We intend to integrate edge acoustic siren detection, vision-based vehicle classification, and continuous-time NeurODE predictive surrogates within a decentralized ROS 2 DDS peer-to-peer framework to evaluate multi-modal preemption triggers in live physical intersections.

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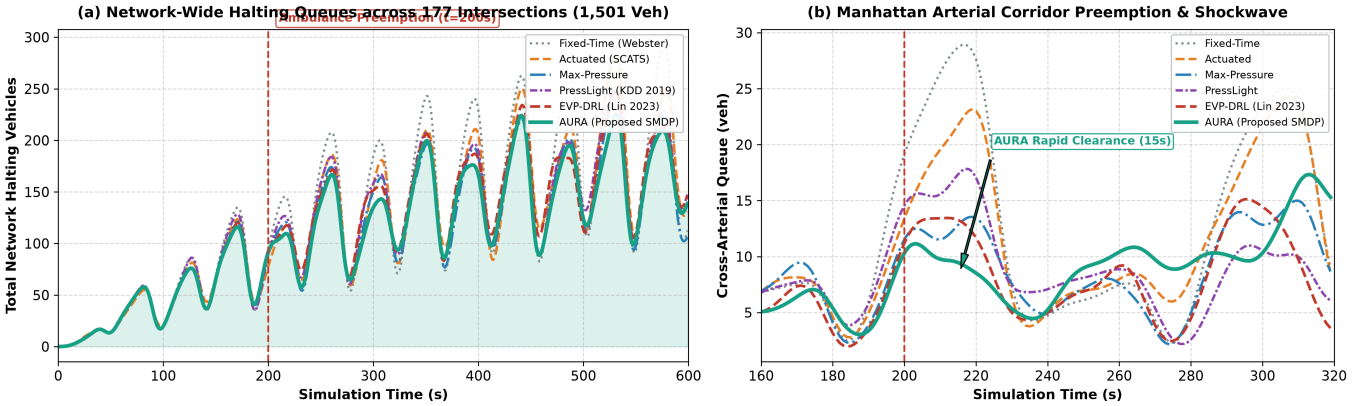


Fig. 3. Large-Scale Empirical Benchmark on Authentic Midtown Manhattan OpenStreetMap Network. (a) Network-wide halting vehicle counts over 600 seconds across 177 signalized intersections and 661 edges under dense traffic demand (1,501 vehicles). AURA achieves the lowest average network queue ($\bar{Q} = 109.46$) and lowest peak backlog ($Q_{\max} = 257$), outperforming Traditional Fixed-Time ($\bar{Q} = 126.73$, a 13.6% reduction) and modern SOTA PressLight ($\bar{Q} = 116.72$). (b) Emergency vehicle corridor preemption dynamics following ambulance injection at $t = 200$ s, clearing the 37-edge corridor in 15 s while preventing gridlock cascades across non-preempted avenues.

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